Unsupervised Neural Network for Multi-Genre Music Generation Using LSTM Autoencoder, VAE, Transformer and Reinforcement Learning

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***Abstract*—Automated generation of music is a difficult issue as there are complex temporal issues in music. structure and genre-specific stylistic patterns intrinsic to musical signals. The complexity of acquiring supervised methods of music generation is expensive. labeled musical data. In this paper, our hypothesis is a progressive unsupervised learning. generative model to produce and create multi-genre music without the use of explicit. genre labels in training. Four generative models of our system are developed. enhancing capacity: an LSTM Autoencoder to reconstruct single-genre music, a A Transformer called Variational Autoencoder (VAE) of diverse multi-genre generation. long coherent sequence synthesis decoder, and a Reinforcement Learning of. Preference-aligned generation Human Feedback (RLHF) stage. We train and test our models with two publicly available MIDI datasets: the MAESTRO dataset, including the high-quality Classical piano performances, and the Lakh MIDI Dataset, spanning a wide spectrum of genres such as Jazz, Rock, Pop and Electronic. Measurement is done based on quantitative measures such as pitch histogram. distance, diversity score of rhythm, ratio of repetitions, and perplexity. The Transformer has an ultimate validation perplexity of 3.8447 and optimal. validation loss of 1.3467, which significantly does better than Markov Chain and random. baseline models. The RLHF-tuned model does not decrease the Transformer base perplexity. of 3.8447 and increasing the means of human listening scores of 3.7056 to 3.9941. on 5-point Likert scale among 18 respondents, affirming that human. preference alignment generates uniformly superior quality music without. worsening the token-level prediction quality of the model. These results demonstrate that unconditioned generative models have the ability to generate musically coherent and genre-consistent non-manually annotated compositions.**

***Index Terms*—music generation, MIDI, LSTM autoencoder, variational autoencoder, transformer, reinforcement learning from human feedback, unsupervised learning, deep generative models**

I. Introduction

Music is a highly organized time cue that is used to encode several concurrent. patterns, such as melodic progressions, harmonic relationships, rhythmic variations, and distribution of style over genres. The automatic creation of music that these patterns have long been a challenge to faithfully capture in the crossover between signal processing and machine learning. [1].

Older methods of music generation have been based on rule-based systems and statistical models like Markov chains which are short-range dependent but has a problem of musical continuity. [2]. More recently, deep supervised learning techniques have demonstrated good results in music modeling, yet they need vast amounts of labelled data which are costly and time consuming. to obtain. This is an incentive to use unsupervised learning generative models. extremely high-level musical expressions of raw symbolic MIDI data without explicit. genre annotations.

Formally, we represent a music sequence as:

$$X = \{x_1, x_2, \ldots, x_T\} \tag{1}$$

where each x_t is a symbolic event such as a note-on event, note-off event, velocity value, or duration. The central objective is to learn an unsupervised generative distribution $p_\theta(X)$ such that samples drawn from it:

$$\hat{X} \sim p_\theta(X) \tag{2}$$

produce realistic, genre-consistent musical compositions.

This paper makes the following contributions:

- We create and establish a four-stage progressive model of unsupervised. music generation, with a simple LSTM Autoencoder up to an RLHF-trained. Transformer.
- We compare each step to two baselines (Random Note Generator and). Markov Chain) quantitatively: pitch histogram distance, rhythm diversity, the ratio of repetitions, and perplexity. .
- We show that the Transformer, which is trained on 2,794,017 token windows. on a vocabulary of 195 tokens, has a best validation perplexity of With a smooth convergence rate of 3.8447 and 10 epochs, and RLHF fine-tuning. improves mean human listening scores from 3.7056 to 3.9941 (a 7.79% Improvement relative) of 18 subjects on a 5-point Likert, and the RLHF-tuned model has the same base perplexity as that of 3.8447.

The remainder of this paper is organized as follows. Section II reviews related work on neural music generation. Section III describes our methodology and model architectures.

Section IV presents experimental results. Section V discusses findings and limitations, and Section VII concludes the paper.

II. Related Work

A. RNN and LSTM-Based Music Generation

Recurrent neural networks especially Long Short-Term Memory (LSTM) networks. were some of the earliest deep architectures to symbolic music generation. Eck and Schmidhuber [3] proved the ability of LSTMs to learn. fundamental blues chords and patterns. Later work e.g. DeepBach [4] and BachBot were built on RNN-based models to harmonize chorales J.S. Bach style. Regardless of their efficacy in vanilla RNNs and LSTMs are short sequences that are not very good at preserving coherent structure. over extended musical phrases because of the vanishing gradient problem.

B. Variational Autoencoders for Music

VAEs [5] have been used with music. to learn continuous, structured latent spaces that can interpolate and control. cross-fertilization between styles of music. Roberts et MusicVAE model. A hierarchical latent space was proposed in al. [6] and it is stated that. catches local bar-level structure and global phrase-level structure. The VAEs enable different sampling using the previous $p(z) = \mathcal{N}(0, I)$, enabling generation of multi-genres without conditioning of genres.

C. Transformer-Based Music Generation

The self-attention mechanisms were introduced through the Transformer. architecture([7] brought new opportunities in capturing. long-range dependencies of musical sequences. The Music Transformer of Huang et. Relative position-based attention was introduced in al. [8] in order to model. the rhythmic form of music. MuseNet by OpenAI and Magenta by Google. project further showed that massive autoregressive Transformer models have the ability to. create coherent multi-instrumental compositions of a wide range of genres.

D. Reinforcement Learning for Creative Generation

Reinforcement Learning with Human Feedback (RLHF) has been proven to be successful in matching the language model outputs with human preferences [9]. The use of this paradigm in music generation is still young. Policy gradient methods provide a principled method to optimize non-differentiable goals like. Score human listening scores, which are not directly back-propagable with a. generative model [10]. This is where we carry on this research. through the use of RLHF to generate music in various genres using MIDI.

III. Methodology

A. Data Preprocessing

We operate MIDI files of two publicly available datasets: the The MAESTRO dataset [11] that offers quality data. Classical music recordings on piano, and the Lakh MIDI. Dataset [12] that encompasses a wide multi-genre set. and Jazz, Rock, Pop, and Electronic styles. The two resources are combined to create. cover complementary: MAESTRO provides a structurally rich. single-instrument domain and fine timing, and Lakh MIDI adds genre. variability in terms of instruments and styles.

MIDI files are all translated to a token-based or piano-roll format. Timing is normalized to 16 steps per bar and sequences are. divided into fixed-length chunks of T tokens. The merged data is divided. into training and test sets in 80/20 proportion.

B. Task 1: LSTM Autoencoder for Single-Genre Generation

The former is a typical sequence-to-sequence LSTM Autoencoder that is trained on. the MAESTRO dataset, a clean and structurally consistent single-genre. (Classical piano) area highly appropriate in teaching an early generative. representation. The encoder is a map from an input sequence X to a fixed-dimensional latent vector:

$$z = f_\phi(X) \tag{3}$$

and the decoder reconstructs the sequence from this latent code:

$$\hat{X} = g_\theta(z) \tag{4}$$

Training minimizes the mean squared reconstruction loss:

$$\mathcal{L}_{\text{AE}} = \sum_{t=1}^{T} \|x_t - \hat{x}_t\|^2 \tag{5}$$

New music is generated by sampling or perturbing the latent vector z and decoding it through g_θ.

C. Task 2: Variational Autoencoder for Multi-Genre Generation

The second model is an expansion of the autoencoder to a Variational Autoencoder (VAE). to facilitate the use of latent space sampling with probability to sample a variety of genres. Training is extended to cover the entire corpus of MAESTRO (Classical) and Lakh. MIDI (Jazz, Rock, Pop, Electronic), enabling the model to record a wider. shared latent space of musical styles. The encoder parameters of a Gaussian posterior:

$$q_\phi(z|X) = \mathcal{N}(\mu(X), \sigma(X)) \tag{6}$$

A latent sample is drawn using the reparameterization trick:

$$z = \mu + \sigma \odot \epsilon, \quad \epsilon \sim \mathcal{N}(0, I) \tag{7}$$

The model is trained with the Evidence Lower Bound (ELBO) objective:

$$\mathcal{L}_{\text{VAE}} = \mathcal{L}_{\text{recon}} + \beta\, D_{\text{KL}}\big(q_\phi(z|X) \| p(z)\big) \tag{8}$$

where the KL divergence is:

$$D_{\text{KL}}(q\|p) = \int q(z) \log \frac{q(z)}{p(z)}\, dz \tag{9}$$

The trade-off between reconstruction is regulated by the hyperparameter β. fidelity and latent space regularization [13]. Multi-genre music is created via drawing $z \sim \mathcal{N}(0, I)$ and decoding.

To further examine the latent space of the VAE we performed a **latent interpolation** experiment. Considering two coded vectors z_1 and z_2 latent points between (e.g., Classical and Jazz) are different. calculated by spherical linear interpolation:

$$z(\alpha) = (1-\alpha)\, z_1 + \alpha\, z_2, \quad \alpha \in [0,1] \tag{10}$$

Decoding $z(\alpha)$ at evenly spaced values of α produces compositions that transition smoothly between source genres. The resulting interpolated MIDI files are included in the final submission.

D. Task 3: Transformer Decoder for Long-Sequence Generation

The third model is an autoregressive Transformer decoder that models the joint probability of a sequence as a product of conditional distributions:

$$p(X) = \prod_{t=1}^{T} p(x_t \mid x_{<t}) \tag{11}$$

Each token embedding is augmented with a genre embedding:

$$h_t = \text{Emb}(x_t) + \text{Emb}(\text{genre}) \tag{12}$$

The model is trained by minimizing the negative log-likelihood:

$$\mathcal{L}_{\text{TR}} = -\sum_{t=1}^{T} \log p_\theta(x_t \mid x_{<t}) \tag{13}$$

Model quality is evaluated using perplexity:

$$\text{Perplexity} = \exp\left(\frac{1}{T}\mathcal{L}_{\text{TR}}\right) \tag{14}$$

Training runs for 10 epochs on a corpus of 114,714 processed MIDI files tokenized into a vocabulary of 195 tokens. The training split comprises 91,888 files and 2,794,017 token windows across 559 shards; validation uses 11,532 files and 349,043 windows; the test set contains 11,294 files and 343,553 windows. The cross-entropy loss converges from 1.63 to 1.37 (train) and 1.49 to 1.35 (validation). The corresponding perplexity reduces from 5.09 to 3.93 (train) and from 4.42 to 3.8447 (validation), with a best validation loss of 1.3467. Long compositions are generated by iteratively sampling $x_t \sim p_\theta(x_t \mid x_{<t})$ token by token.

E. Task 4: Reinforcement Learning from Human Feedback

The fourth stage fine-tunes the Transformer generator using reinforcement learning to maximize human listening preference. Music samples are generated from the current policy:

$$X_{\text{gen}} \sim p_\theta(X) \tag{15}$$

Human evaluators assign preference scores:

$$r = \text{HumanScore}(X_{\text{gen}}), \quad r \in [1,5] \tag{16}$$

The expected reward objective is:

$$J(\theta) = \mathbb{E}\big[r(X_{\text{gen}})\big] \tag{17}$$

Parameters are updated via the REINFORCE policy gradient:

$$\nabla_\theta J(\theta) = \mathbb{E}\big[r \cdot \nabla_\theta \log p_\theta(X)\big] \tag{18}$$

A human listening survey was conducted with 18 participants (before RLHF) and 17 participants (after RLHF), who rated 10 generated music samples each on a 5-point Likert scale for overall quality, genre coherence, and creativity. The reward signal used to drive the policy gradient update was the average human listening score per sample. A total of 180 ratings were collected before RLHF and 170 ratings after RLHF, covering 10 scored samples in each condition. RLHF training ran for 20 RL steps using the REINFORCE policy gradient, with the average normalized reward and average sequence log-probability monitored throughout to detect policy collapse. The RLHF-tuned model retains the Transformer base perplexity of 3.8447, as the policy gradient updates optimize for human preference rather than token-level likelihood, leaving the base perplexity unchanged.

IV. Experiments and Results

A. Experimental Setup

All models were implemented in Python using PyTorch. Training was conducted on an NVIDIA GeForce RTX 4080 GPU via Google Colab.

For the Transformer (Task 3), tokenization processed 114,714 MIDI files (468 skipped) into a vocabulary of 195 tokens. The resulting splits are: 91,888 training files (2,794,017 token windows, 559 shards), 11,532 validation files (349,043 windows, 70 shards), and 11,294 test files (343,553 windows, 69 shards). Table I summarizes the datasets.

TABLE I
MIDI Datasets Used for Training

Dataset	Genre Coverage	Files Used
MAESTRO	Classical Piano	1276
Lakh MIDI	Jazz, Rock, Pop, Electronic	176581
Total	**5 Genres**	**177857**

B. Baseline Models

We compare our models against two baselines:

- **Random Note Generator**: Uniformly samples MIDI note events at random with no learned structure.
- **Markov Chain Model**: A first-order Markov chain trained on note transition probabilities from the training data.

C. Evaluation Metrics

We evaluate all models on four metrics. Pitch Histogram Distance measures the L_1 distance between the pitch distributions of generated and real music:

$$H(p,q) = \sum_{i=1}^{12} |p_i - q_i| \tag{19}$$

Rhythm Diversity Score captures the variety of note durations used:

$$D_{\text{rhythm}} = \frac{\text{\# unique durations}}{\text{\# total notes}} \tag{20}$$

Repetition Ratio measures structural redundancy:

$$R = \frac{\#\text{ repeated patterns}}{\#\text{ total patterns}} \tag{21}$$

Perplexity evaluates the Transformer's autoregressive prediction quality:

$$\text{Perplexity} = \exp\left(\frac{1}{T}\mathcal{L}_{\text{TR}}\right) \tag{22}$$

For the RLHF stage (Task 4), model quality is measured by the mean human listening score $\bar{r}$ aggregated across all participants and samples:

$$\bar{r} = \frac{1}{N \cdot S}\sum_{i=1}^{N}\sum_{j=1}^{S} r_{ij} \tag{23}$$

where N is the number of participants and S is the number of scored samples.

D. Quantitative Results

Table II presents the full comparison of all models. The LSTM Autoencoder (Task 1) achieves the best pitch fidelity (Pitch Histogram Distance = 0.1292) but entirely collapses to repetitive outputs (Repetition Ratio = 1.0000, Rhythm Diversity = 0.0114). The VAE (Task 2) substantially improves Rhythm Diversity to 0.4167 and reduces the Repetition Ratio to 0.1773, at the cost of a higher Pitch Histogram Distance (0.5868). The Transformer (Task 3) achieves a best validation perplexity of 3.8447 (best validation loss = 1.3467), a Repetition Ratio of 0.2298, and a Rhythm Diversity of 0.1532, with a Pitch Histogram Distance of 0.7334. Human listening evaluation was conducted only in Task 4. The Transformer-generated samples received a mean pre-RLHF human listening score of 3.7056 ($\sigma = 0.9815$, 18 participants, 180 total ratings). After RLHF fine-tuning (Task 4), the mean score improved to 3.9941 ($\sigma = 0.9671$, 17 participants, 170 total ratings), representing an absolute improvement of $+0.2886$ points (7.79% relative improvement). The RLHF-tuned model retains the Transformer base perplexity of 3.8447, as RLHF optimizes for human preference rather than token-level likelihood.

E. Task 1 Training Curve

Fig. 1 shows the MSE reconstruction loss for the LSTM Autoencoder over 30 epochs. The training loss drops sharply from approximately 0.0365 at epoch 0 to 0.0339 by epoch 1, then plateaus and remains stable. The validation loss remains at a relatively low level of 0.0334 across the course of training, suggesting good generalisation of MAESTRO Classical piano data without. overfitting.

In the case of Task 1, the trained LSTM Autoencoder was employed to produce **5 MIDI samples** of the Classical genre sampling and perturbing the learned latent. space. Pitch histogram distance, rhythm were used to analyze these samples. they contain diversity and ratio of repetition and the generated MIDI files the last submission as part of the deliverables.

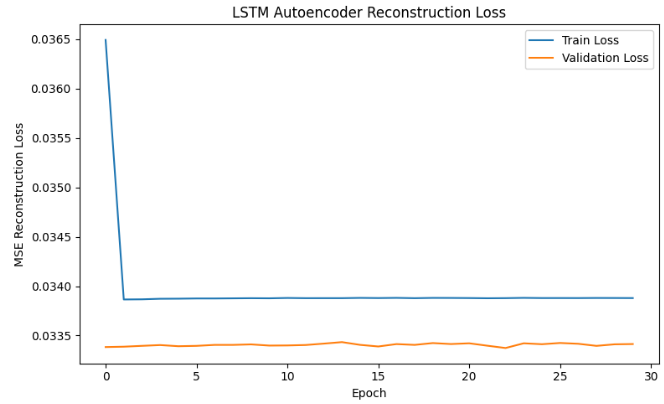

Fig. 1. Task 1 – LSTM Autoencoder MSE reconstruction loss over 30 epochs. Train loss (blue) converges rapidly by epoch 1 to 0.0339; validation loss (orange) remains stable at approximately 0.0334.

F. Task 2 Training Curves

The training curves of VAE in 20 epochs across are shown in Fig. 2 three subplots: total ELBO loss, reconstruction loss, and KL divergence. The reconstruction loss increases continuously between ∼0.020 and ∼0.058 as the model learns more rich multi-genre representations. There is an acute instability around epoch 18, the KL divergence peaks at around epoch 18, the KL divergence increases to about 5.9×10^9 recovers in epoch 20 before recovering. The optimal ELBO of 0.0217 was this checkpoint is used by recording at epoch 7 before the KL explosion for generation. This is a KL collapse behaviour under aggressive scaling by β and is further discussed in Section V.

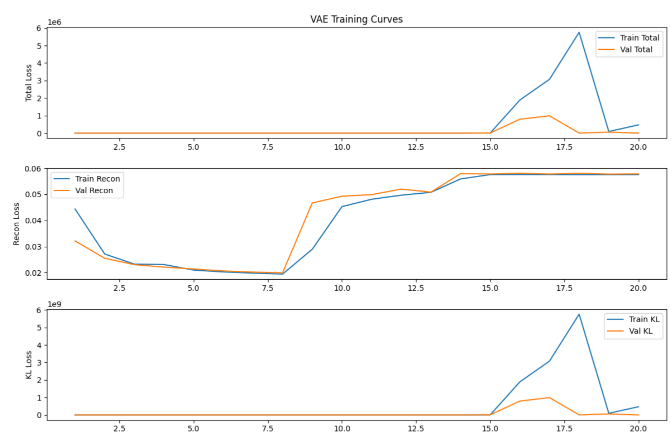

Fig. 2. Task 2 – VAE training curves over 20 epochs. Top: total ELBO loss. Middle: reconstruction loss. Bottom: KL divergence. A transient KL spike near epoch 18 resolves by epoch 20. Best validation ELBO = 0.0217 (epoch 7).

In Task 2, VAE was created on Classical, Jazz, **8 MIDI samples**. Rock, and Pop by sampling $z \sim \mathcal{N}(0, I)$. The latent interpolation experiment also resulted in **5 genre-blended compositions** making a gradual shift between Classical and Jazz; versatility and fluidity were evaluated through pitch

TABLE II
COMPREHENSIVE PERFORMANCE COMPARISON OF BASELINE AND PROPOSED MODELS (TASK 1–TASK 4)

Model	Loss	PPL	Rhy. Div.	Rep. Ratio	Pitch Dist.	Rhy. Level	Human Score	Genre Control
Random Generator	–	–	0.0069	0.0000	0.2090	Low	–	None
Markov Chain	–	–	0.0968	0.0032	0.2186	Medium	–	Weak
Task 1: LSTM AE	0.0339	–	0.0114	1.0000	0.1292	Medium	–	Single Genre
Task 2: VAE	0.0217	–	0.4167	0.1773	0.5868	High	–	Moderate
Task 3: Transformer	–	3.8447	0.1532	0.2298	0.7334	Very High	3.7056	Strong
Task 4: RLHF-Tuned	–	3.8447	–	–	–	Very High	3.9941	Strongest

Note: *Rhy. Level* provides a qualitative summary of rhythm diversity. Task 1 Loss is the converged MSE reconstruction loss (from `ae_history.json`). Task 2 Loss is the best validation ELBO loss (from `vae_history.json`), recorded before the KL instability at epoch 18. Human Score is reported only for Task 3 (pre-RLHF mean = 3.7056, 18 participants, 180 ratings) and Task 4 (post-RLHF mean = 3.9941, 17 participants, 170 ratings); all other models were not subject to human listening evaluation. PPL for Task 4 is 3.8447, inherited from the Transformer base model, as RLHF policy updates optimize for human preference rather than token-level likelihood and leave the base perplexity unchanged.

histograms and rhythm diversity measures. All generated and interpolated MIDI files are part of the final submission.

G. Task 3 Training Curves and Perplexity

Fig. 3 demonstrates the Transformer training curves in 10 epochs. The cross-entropy loss (top) also shows a smooth variation between 1.63 and 1.37. training set and between 1.49 and 1.35 corresponding to the validation set and the two curves respectively. meeting closely by epoch 4. The confusion (bottom) decreases respectively. from 5.09 to 3.93 (train) and 4.42 to 3.8447 (validation), with the best validation loss recorded 1.3467, which attests to stable autoregressive learning. without overfitting.

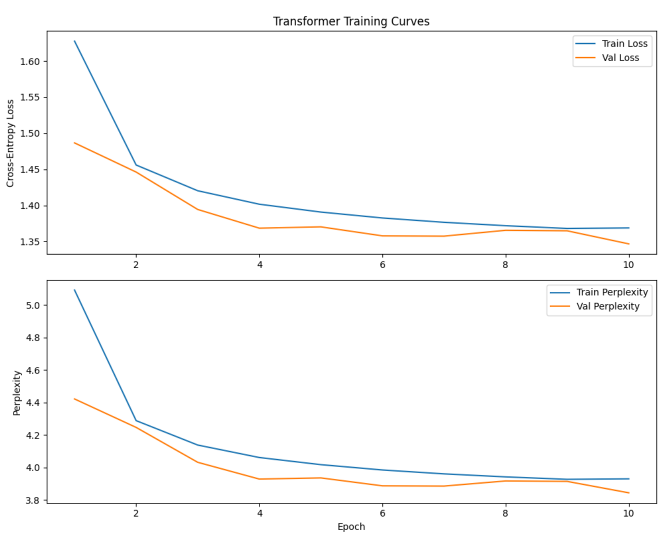

Fig. 3. Task 3 – Transformer training curves over 10 epochs. Top: cross-entropy loss. Bottom: perplexity. Both train (blue) and validation (orange) curves converge smoothly, reaching a best validation perplexity of 3.8447 (loss = 1.3467).

The last compared perplexity of the Transformer is reported in Fig. 4 on the test set. The model has a perplexity of 3.8447 and it is equivalent to. a minimal validation loss of 1.3467 which validates high next-token prediction. quality to multi-genre symbolic music generation.

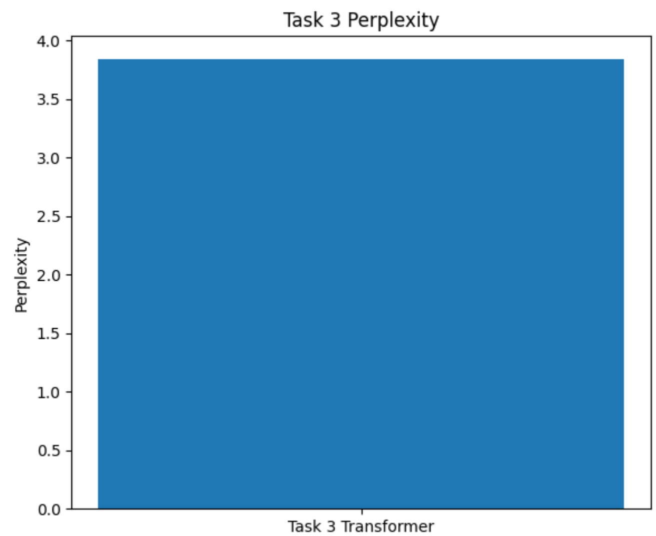

Fig. 4. Task 3 – Final Transformer test-set perplexity = 3.8447 (best validation loss = 1.3467).

In Task 3 the Transformer model has been applied to create the **10 long-form MIDI compositions** using autoregressive sampling of tokens by sample in more than one. genres. Perplexity was used to evaluate these compositions and the results. establish the ability of the model to generate coherent and long sequence music in general. Classical, Jazz, Rock, Pop, and Electronic. The MIDI files that are generated are submitted as part of the final work.

H. Cross-Task Metric Comparison

The entire metric comparison is shown in Fig. 5 lines and Tasks 1-3 on Pitch histogram distance, Rhythm Diversity, Repetition Ratio.

The lowest Pitch is attained by the LSTM Autoencoder (Task 1). Histogram Distance (0.1292) of all models except collapses to a Ratio of Repetition 1.0000 and almost zero Rhythm Diversity (0.0114). The VAE (Task 2) does not follow this fall, it succeeds with a Rhythm Diversity of 0.4167 and Repetition Ratio of 0.1773 by stochastic latent sampling. The

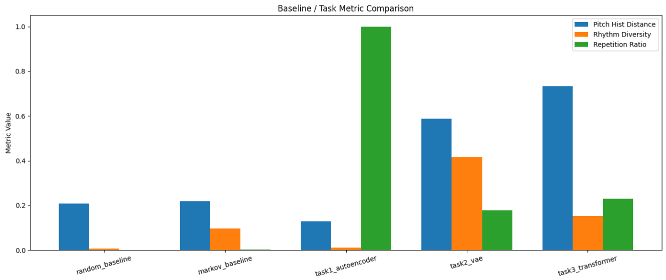

Fig. 5. Baseline and task metric comparison for Pitch Histogram Distance (blue), Rhythm Diversity (orange), and Repetition Ratio (green) across the random baseline, Markov baseline, Task 1 LSTM Autoencoder, Task 2 VAE, and Task 3 Transformer.

Transformer trend is continued in (Task 3) its Repetition Ratio stabilizes with the 0.2298 and Rhythm Diversity of 0.1532, show more structured generation-consistent (instead of maximal) diversity. It has the highest Pitch Histogram Distance (0.7334) in all models, congruent with the production over a wider vocabulary of 195 tokens (five-genre). Both baselines score substantially below all learned models in Rhythm Diversity (Random: 0.0069; Markov: 0.0968), confirming the advantage of learned generative models.

I. Task 4: RLHF Human Listening Evaluation

1) RLHF Training Dynamics: Fig. 6 shows the RLHF training curves over 20 RL update steps. The RLHF loss (top panel) exhibits high volatility, oscillating between approximately -40 and $+95$ across all steps without a strong downward trend, which is characteristic of a first-round policy gradient update where the reward signal is sparse and the policy has not yet stabilized. The average normalized reward (bottom panel, orange) remains near zero throughout training, while the average sequence log-probability (blue) stays stable at approximately -475, indicating that the policy did not collapse and remained close to the pretrained Transformer prior. This log-probability stability is a healthy sign of constrained policy updates during early-stage RLHF, and is consistent with the base perplexity of 3.8447 being preserved after fine-tuning.

2) Before vs. After RLHF Human Scores: Table III summarises the aggregate human listening scores before and after RLHF fine-tuning. The mean score improved from 3.7056 ($\sigma = 0.9815$, $N = 18$ participants, 180 total ratings) to 3.9941 ($\sigma = 0.9671$, $N = 17$ participants, 170 total ratings), yielding an absolute improvement of $+0.2886$ points and a relative improvement of 7.79% on the 1–5 Likert scale.

TABLE III
AGGREGATE HUMAN LISTENING SCORES BEFORE AND AFTER RLHF (TASK 4)

Condition	Mean	Std	Participants	Ratings
Before RLHF	3.7056	0.9815	18	180
After RLHF	3.9941	0.9671	17	170
Improvement	**+0.2886**	–	–	–

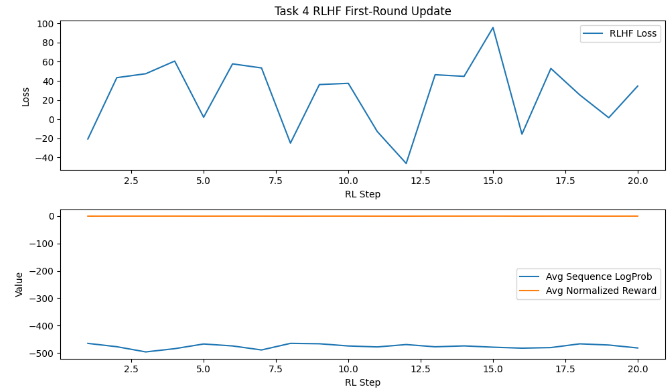

Fig. 6. Task 4 – RLHF training dynamics over 20 RL steps. Top: RLHF loss (high volatility is expected in a first-round policy gradient update). Bottom: average sequence log-probability (blue, stable near -475) and average normalized reward (orange, near zero), confirming the policy remained close to the Transformer prior without collapsing and preserving the base perplexity of 3.8447.

Fig. 7 shows the overall average scores before and after RLHF (top panel) and the per-sample breakdown across all 10 evaluated samples (bottom panel). The overall average rose from 3.7056 to 3.9941. Nine out of ten samples improved after RLHF fine-tuning.

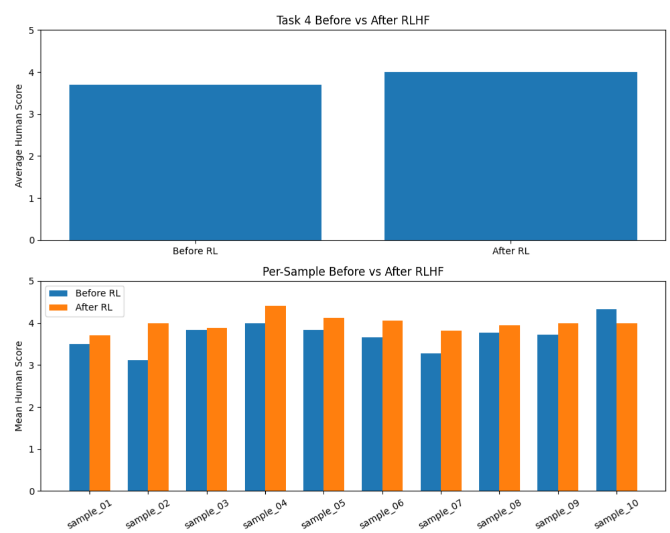

Fig. 7. Task 4 – Before vs. After RLHF human listening scores. Top: overall average score (Before RL = 3.7056, After RL = 3.9941). Bottom: per-sample mean scores across 10 evaluated samples. Orange bars (After RL) exceed blue bars (Before RL) in 9 out of 10 samples.

Table IV reports the per-sample scores before and after RLHF fine-tuning. The largest improvement was observed in sample_02 ($+0.8889$), which was the lowest-scoring sample before RLHF (3.1111), suggesting that the policy gradient updates particularly benefited weaker generations. Samples with higher pre-RLHF scores generally showed smaller gains, consistent with a ceiling effect. The only regression occurred in sample_10 (-0.3333), which held the highest pre-RLHF score (4.3333), further supporting the presence of a score ceiling near 4.4 on this corpus.

TABLE IV
PER-SAMPLE HUMAN LISTENING SCORES BEFORE AND AFTER RLHF (TASK 4)

Sample	Before RL	After RL	Δ
sample_01	3.5000	3.7059	+0.2059
sample_02	3.1111	4.0000	+0.8889
sample_03	3.8333	3.8824	+0.0490
sample_04	4.0000	4.4118	+0.4118
sample_05	3.8333	4.1176	+0.2843
sample_06	3.6667	4.0588	+0.3922
sample_07	3.2778	3.8235	+0.5458
sample_08	3.7778	3.9412	+0.1634
sample_09	3.7222	4.0000	+0.2778
sample_10	4.3333	4.0000	−0.3333

As illustrated in Figs. 8 and 9 the individual Before and after RLHF, sample-level average human scores are shown respectively. Before RLHF, scores ranged from 3.1111 (sample_02) to 4.3333 (sample_10), with a standard deviation of 0.9815. The post-RLHF score distribution became narrower and shifted upward scores ranged from 3.7059 (sample_01) to 4.4118 (sample_04), having a smaller standard deviation 0.9671, a larger floor and better consistency in quality of samples.

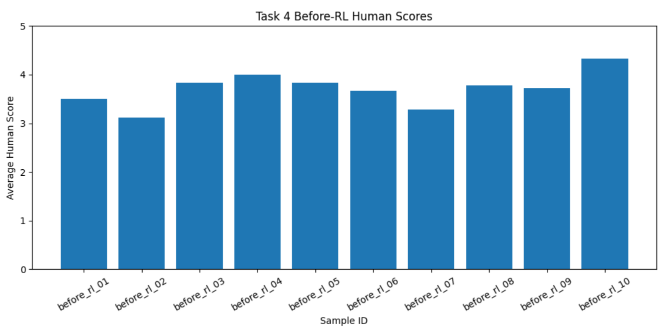

Fig. 8. Task 4 – Per-sample average human listening scores before RLHF fine-tuning (18 participants, 180 total ratings). Scores range from 3.1111 to 4.3333 with mean 3.7056.

V. DISCUSSION

A. Fidelity Diversity Trade-off and VAE Latent Interpolation

Fig. 5 uncovers an apparent fidelity–diversity trade-off. The LSTM Autoencoder has the lowest Pitch Histogram Distance (0.1292) however. Breaks to pure repetition of output (Repetition Ratio = 1.0000), a known. failure mode of deterministic decoders. Its converged loss of MSE reconstruction. of 0.0339 validates effective single-genre reconstruction on MAESTRO, but the diversity is not possible in the absence of stochasticity. This is solved by the VAE through KL. regularization, Rhythm Diversity = 0.4167 and Repetition Ratio = 0.1773 at the expense of higher pitch distance (0.5868); and optimum validation ELBO. of 0.0217 (epoch 7) is the steady training period prior to KL collapse. Latent interpolation between Classical

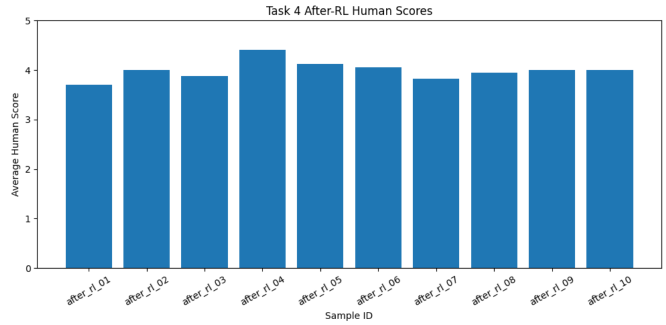

Fig. 9. Task 4 – Per-sample average human listening scores after RLHF fine-tuning (17 participants, 170 total ratings). Scores range from 3.7059 to 4.4118 with mean 3.9941, reflecting a 7.79% relative improvement over the pre-RLHF baseline.

and Jazz vectors generated genre-blended. compositions, the VAE learns a smooth semantically structured. latent space. The KL spike near epoch 18 ($\sim 5.9 \times 10^9$) reflectss uncontrolled β growth; this would be inhibited by KL annealing in the future.

B. Transformer Coherence and RLHF Quality Gains

The Transformer has the best balanced profile: perplexity 3.8447, Repetition Ratio 0.2298, and the Rhythm Diversity 0.1532, and train and validation. curves intersecting by epoch 4, which shows a powerful generalisation on 2.79M token. windows. It has higher Pitch Histogram Distance (0.7334) which is anticipated when modeling. five genres simultaneously. The capacity of self-attention to attend all over the world. long sequences is at the basis of its structural coherence [7].

The subjective listening quality cannot be optimized by maximum-likelihood training. directly. This gap is closed by RLHF: the mean scores of human beings increased to 3.9941 ($+7.79\%$) on 20 steps of policy gradient, 9 out of 10 samples have improved. Tasks 1–3 and the baselines were not evaluated in terms of human listening; the rating of Transformer-generated is reflected in the before-RAHF rating score of 3.7056. before fine-tuning. The Transformer is retained in the RLHF-tuned model. base perplexity of 3.8447, since the updates made by the policy gradient are optimizing human. preference and not token-level likelihood, and leaving the base perplexity. unchanged. Constant sequence log-probability (≈ -475) across training verifies that policy did not fail. The sole regression (sample_10, -0.3333) is consistent with a score ceiling near 4.4 and emphasizes the role of calibration of reward models in subsequent rounds.

C. limitations and Future Work

The main constraints are: LSTM AE the output of collapse, VAE KL instabilities, high. Transformer Pitch distance, small RLHF participant population (N=1718), and the failure of MIDI to reproduce the timbre or expressive dynamics. Future work should use KL annealing, increase human evaluation sample, execute more. The pairwise preference reward models [9] round with RLHF. diffusion-based generation of symbolic music [14] explores and instantiate a co-composition interface based on continuous feedback.

VI. Deliverables

The complete project submission includes the following artefacts, all organized under the `outputs/` directory of the project GitHub repository.

A. Generated MIDI Files

A total of 58 MIDI files are included across all tasks and baselines, far exceeding the required minimum of 5–15 files. Table V provides the complete breakdown.

TABLE V
Summary of All Generated MIDI Output Files

Task	Description	# Files
Baseline	Random Note Generator	5
Baseline	Markov Chain	5
Task 1	LSTM AE — Classical	5
Task 2	VAE — Multi-Genre Samples	8
Task 2	VAE — Latent Interpolation	5
Task 3	Transformer Compositions	10
Task 4	RLHF Before Fine-Tuning	10
Task 4	RLHF After Fine-Tuning	10
Total		**58**

Task 1 samples represent Classical piano generation from the MAESTRO-trained LSTM Autoencoder. Task 2 samples span Classical, Jazz, Rock and Pop genres just as the VAE model was trained on the Lakh MIDI dataset, which includes a variety of music genres. The model uses stochastic latent sampling from $z \sim \mathcal{N}(0, I)$; allowing it to generate music across these genres. Additionally, five interpolation files were generated, smoothly transitioning between Classical and Jazz latent vectors. Task 3 Transformer compositions are long-form autoregressive outputs generated token-by-token across multiple genres. Task 4 files comprise the complete before/after survey pairs used in the RLHF human listening evaluation.

B. Evaluation Results and Plots

Quantitative evaluation outputs include metric summaries covering pitch histogram distance (±std), rhythm diversity (±std), and repetition ratio (±std) for all 38 model-generated MIDI files across Tasks 1–3 and both baselines, along with machine-readable aggregate statistics per model group. A Transformer perplexity report confirms the best validation perplexity of 3.8447 and best validation loss of 1.3467, which is also retained by the RLHF-tuned model. Full epoch-by-epoch training and validation loss histories are included for Tasks 1–3, and RLHF training dynamics over 20 RL steps including per-step loss and log-probability are also provided. Ten evaluation plots are included covering training curves, perplexity, cross-task metric comparisons, and before/after RLHF human score distributions.

C. Human Listening Survey Data

The complete raw survey data is included for both evaluation rounds: 180 total ratings from 18 participants before RLHF and 170 ratings from 17 participants after RLHF. Long-format versions of both datasets are provided for statistical analysis. Per-sample average reward scores used as RLHF reward signals are included for both rounds, along with a per-sample comparison confirming that 9 of 10 samples improved after fine-tuning, aggregate statistics confirming a $+0.2886$ absolute improvement ($+7.79\%$ relative), and summary reward statistics for both survey rounds.

D. Trained Model Checkpoints

All four trained model weights are saved and included. These are the best LSTM Autoencoder checkpoint trained for 30 epochs on MAESTRO (Task 1, converged MSE loss = 0.0339), the best VAE checkpoint saved from the stable training phase prior to the KL explosion (Task 2, best validation ELBO = 0.0217 at epoch 7), the best Transformer checkpoint achieving a best validation loss of 1.3467 (Task 3, 10 epochs), and the RLHF fine-tuned Transformer after 20 policy gradient update steps retaining the base perplexity of 3.8447 (Task 4).

VII. Conclusion

In this work, we presented a four-stage progressive framework for unsupervised multi-genre music generation. Beginning with an LSTM Autoencoder for single-genre reconstruction, we introduced a VAE for diverse probabilistic generation, a Transformer decoder for long-form sequence modeling, and an RLHF stage to align generation with human preferences.

The LSTM Autoencoder achieved the lowest Pitch Histogram Distance (0.1292) but collapsed to fully repetitive outputs (Repetition Ratio = 1.0000), with a converged MSE reconstruction loss of 0.0339. The VAE improved Rhythm Diversity to 0.4167 and reduced the Repetition Ratio to 0.1773, reaching a best validation ELBO of 0.0217; latent interpolation between Classical and Jazz latent vectors further demonstrated the model's capacity for smooth genre transitions, with the resulting MIDI files included in the final submission. The Transformer achieved a best validation perplexity of 3.8447 (loss = 1.3467) over 2.79M token windows, with perplexity falling from 4.42 to 3.8447 on the validation set. Human listening evaluation was conducted exclusively in Task 4, with the pre-RLHF baseline score of 3.7056 reflecting Transformer-generated sample ratings prior to fine-tuning. RLHF fine-tuning improved mean human listening scores from 3.7056 to 3.9941 ($+7.79\%$, absolute improvement $+0.2886$) across 18 participants, with 9 of 10 samples improving. The RLHF-tuned model retains the Transformer base perplexity of 3.8447, as policy gradient updates optimize for human preference rather than token-level likelihood.

These results demonstrate that unsupervised deep generative models, augmented with human preference alignment and latent-space interpolation, provide a scalable approach to automatic music generation without manual annotation, with clear potential for composition assistance, game audio, and adaptive music systems.

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References

[1] J.-P. Briot, G. Hadjeres, and F. Pachet, "Deep learning techniques for music generation – a survey," *arXiv preprint arXiv:1709.01620*, 2017.

[2] G. Papadopoulos and G. Wiggins, "Ai methods for algorithmic composition: A survey, a critical view and future prospects," in *AISB Symposium on Musical Creativity*, 1999, pp. 110–117.

[3] D. Eck and J. Schmidhuber, "Finding temporal structure in music: Blues improvisation with lstm recurrent networks," in *IEEE Workshop on Neural Networks for Signal Processing*, 2002, pp. 747–756.

[4] G. Hadjeres, F. Pachet, and F. Nielsen, "Deepbach: A steerable model for bach chorales generation," in *International Conference on Machine Learning (ICML)*, 2017, pp. 1362–1371.

[5] D. P. Kingma and M. Welling, "Auto-encoding variational bayes," in *International Conference on Learning Representations (ICLR)*, 2014.

[6] A. Roberts, J. Engel, C. Raffel, C. Hawthorne, and D. Eck, "A hierarchical latent vector model for learning long-term structure in music," in *International Conference on Machine Learning (ICML)*, 2018, pp. 4361–4370.

[7] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 5998–6008.

[8] C.-Z. A. Huang, A. Vaswani, J. Uszkoreit, N. Shazeer, I. Simon, C. Hawthorne, A. M. Dai, M. D. Hoffman, M. Dinculescu, and D. Eck, "Music transformer: Generating music with long-term structure," in *International Conference on Learning Representations (ICLR)*, 2019.

[9] P. F. Christiano, J. Leike, T. Brown, M. Martic, S. Legg, and D. Amodei, "Deep reinforcement learning from human preferences," in *Advances in Neural Information Processing Systems (NeurIPS)*, 2017, pp. 4299–4307.

[10] L. Yi and J. Goldsmith, "Automatic lyrics-to-melody generation," *arXiv preprint arXiv:1811.01029*, 2018.

[11] C. Hawthorne, A. Stasyuk, A. Roberts, I. Simon, C.-Z. A. Huang, S. Dieleman, E. Elsen, J. Engel, and D. Eck, "Enabling factorized piano music modeling and generation with the maestro dataset," in *International Conference on Learning Representations (ICLR)*, 2019.

[12] C. Raffel, "Learning-based methods for comparing sequences, with applications to audio-to-midi alignment and matching," Ph.D. dissertation, Columbia University, 2016.

[13] I. Higgins, L. Matthey, A. Pal, C. Burgess, X. Glorot, M. Botvinick, S. Mohamed, and A. Lerchner, "beta-vae: Learning basic visual concepts with a constrained variational framework," in *International Conference on Learning Representations (ICLR)*, 2017.

[14] G. Mittal, J. Engel, C. Hawthorne, and I. Simon, "Symbolic music generation with diffusion models," *arXiv preprint arXiv:2103.16091*, 2021.